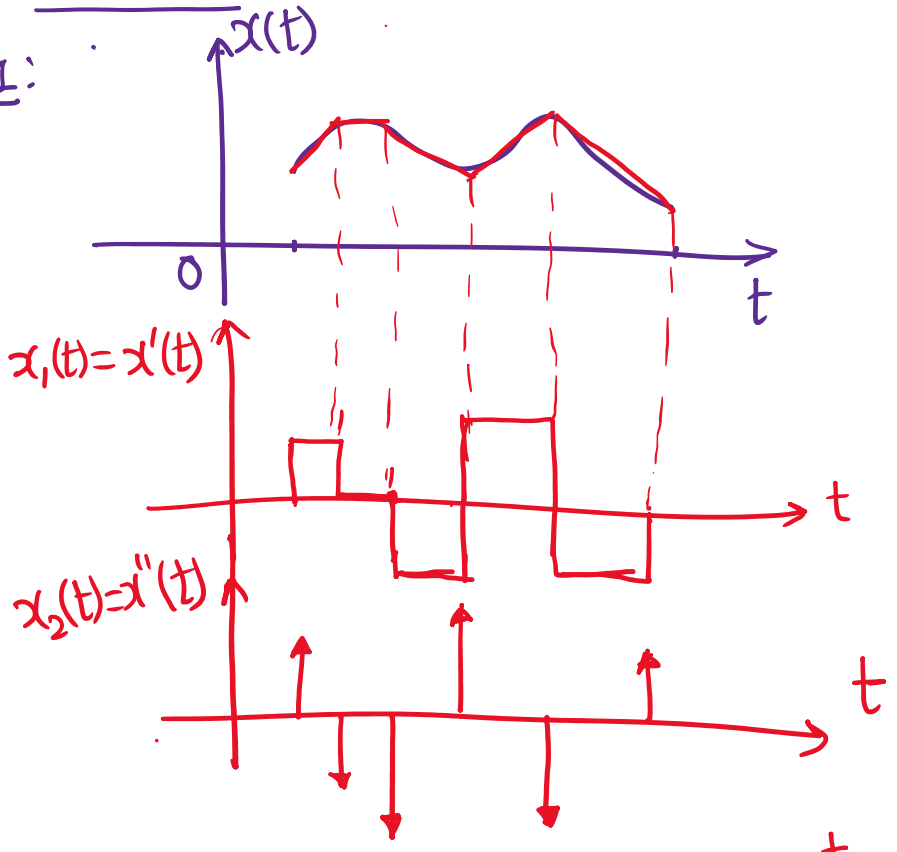
C.T.F.T.:

Ex.1:



$X(\omega) =$

$$X_1(\omega) = \frac{X_2(\omega)}{j\omega} + \pi\delta(\omega)X_2(0)$$

$X_2(\omega)$: easily calculated (using F.T. of impulses).

$$X_1(\omega) = \frac{X_2(\omega)}{j\omega} + \pi\delta(\omega)X_2(0) \Rightarrow x_1(t) = \int_{-\infty}^{t} x_2(\tau)\,d\tau$$

$$\therefore x_1(\infty) = \int_{-\infty}^{\infty} x_2(\tau)\,d\tau = 0$$

$x(t)$ is finite durat^n signal.

$$X_2(0) = \int_{-\infty}^{\infty} x_2(t)\,dt = 0$$

$$\therefore X_1(\omega) = \frac{X_2(\omega)}{j\omega}$$

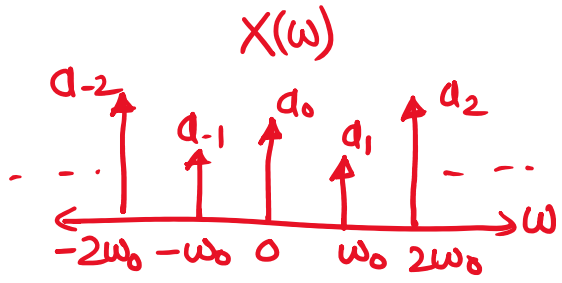
Similarly, $X(\omega) = \dfrac{X_1(\omega)}{j\omega} + \overset{0}{\pi\delta(\omega)X_1(0)} = \dfrac{X_1(\omega)}{j\omega} = \dfrac{-X_2(\omega)}{\omega^2}$

Ex.2: $x(t)$ is periodic $\longleftrightarrow$ $X(\omega)$?

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad \Big\}\ \text{C.T.F.S.}$$

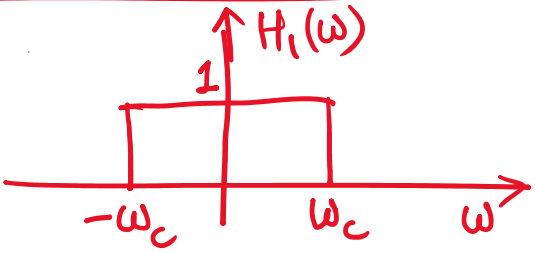
$$1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$\therefore X(\omega) = \sum_{k=-\infty}^{\infty} a_k\, 2\pi\,\delta(\omega - k\omega_0)$$



$k = -\infty$

Ex.3



$$x(t) \longrightarrow \boxed{h_1} \longrightarrow y(t)$$

Inverse system of h_1?

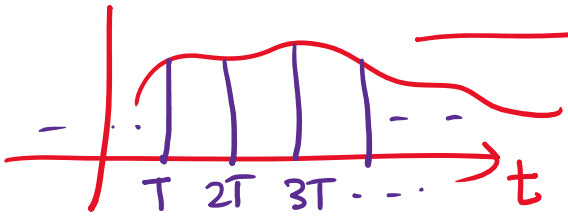
$$x(t) \longrightarrow \boxed{h_1} \longrightarrow \overset{\text{Inverse}}{\boxed{h_2}} \longrightarrow x(t)$$

$\Rightarrow \quad H_1(\omega)\,H_2(\omega) = 1 \Rightarrow H_2(\omega) = \dfrac{1}{H_1(\omega)} \Rightarrow$ Undefined

$\therefore$ Non-invertible

Sampling Theorem:

$x(t)$



$\longrightarrow x[n]$?